

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 03

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark.</i> • The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) (1)	1
01.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Light: UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions (1) • Temperature: higher temperatures increase the rate of chemical breakdown / microbial decomposition (1) • Soil microbes / oxygen: aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils (1) • pH / presence of other chemicals may increase or decrease the rate of degradation (1) • Adsorption onto clay / organic matter may protect from degradation and increase persistence (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
02.1	<i>1 mark for the correct answer.</i> • $(120 + 95) - (2 + 45) = 168 \text{ t ha}^{-1}$ (1)	1
02.2	<i>1 mark for each point.</i> • Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO₂ (1) • Harvest removes biomass so less dead organic matter is returned; pasture roots and litter constantly add organic matter (1)	2
02.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Low / zero tillage (1) • Adding manure, compost or mulch (1) • Growing cover crops / leaving crop residues (1) • Converting to long-term grass or tree crops in rotation (1)	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	
03.1	<i>1 mark for each point.</i> • Spearman's rank correlation coefficient (1) • Chi-squared test (frequencies / counts in categories) (1) • t-test (two means of measured data) (1) • Spearman's rank correlation coefficient (1)	4
03.2	<i>1 mark.</i> • Chi-squared compares frequencies / counts in categories; it cannot be used for continuously variable measured data (1)	1
03.3	<i>1 mark for each point.</i> • The value (ignoring sign) exceeds the critical value so the correlation is significant at the 5% level (1) • The negative sign shows a negative correlation: one variable decreases as the other increases (1) • There is less than a 5% probability the correlation arose by chance; reject the null hypothesis (but correlation is not proof of causation) (1)	3
03.4	<i>1 mark for each point.</i> • Show the standard deviation / range of the original values around each mean (1) • If the bars of two means overlap greatly the difference is probably not significant; little overlap suggests a significant difference (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 422-423; AQA spec 3.7.2.3.	
04.1	<i>1 mark for the correct answer.</i> • $4.2 / 0.7 = 6$ (1)	1
04.2	<i>1 mark for each point.</i> • Long summer days (almost 24 hours of daylight) so the total energy received is high, even though the Sun is low (1) • In winter days are very short / the Sun does not rise, so light levels are extremely low (1)	2
04.3	<i>1 mark for each point.</i> • Reflectors can only focus parallel rays of direct sunlight onto the absorber (1) • Clouds, smoke and dust scatter the light so it cannot be focused and intensified (1)	2
04.4	<i>1 mark.</i> • Very large areas of collectors are needed to harness significant amounts of energy (and the changing angle of incidence reduces it further) (1)	1

Question	Marking guidance	Marks
04.5	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Problems: intermittency (daily and seasonal cycles), unpredictable cloud cover, low energy density, changing angle of incidence, land requirements, water for cleaning in deserts, toxic wastes in manufacture (silicon tetrachloride, cadmium), efficiency loss at high temperature • Photothermal: evacuated tubes reduce conduction / convection losses; thermal stores (water, sand, concrete, molten salt) allow later use; passive solar architecture, brise soleil, light tubes; heat pumps deliver up to four times the energy used • Photovoltaic developments: multijunction cells absorbing more wavelengths, anti-reflective / moth-eye textured surfaces (smooth cells reflect 30%), PV-thermal hybrids cooling the cell, transparent PV in windows, heliostats keeping the optimum angle, self-cleaning nanohydrophobic surfaces • Concentrating solar power with molten-salt storage to 550 °C so steam turbines run when needed - overcomes intermittency • Integration: grid-connected solar farms, rooftop space avoiding land conflicts, batteries, pumped storage, links to energy storage and demand management • Evaluation: strong in sunny deserts / low latitudes, weaker in high latitudes in winter; cost falling; manufacturing impacts; no single technology solves all problems; justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	
05.1	<i>Correct answer only.</i> • B - tiny pores between clay particles allow capillary rise; none in sand (1)	1
05.2	<i>1 mark for each point.</i> • Hygroscopic: held tightly by adhesive forces on particle surfaces; cannot drain and cannot be absorbed by plants (1) • Capillary: held in pores by surface tension; does not drain but can be absorbed by plants (1) • Gravitational: drains freely through large pores; available to plants only briefly as it passes (1)	3
05.3	<i>1 mark.</i> • Small round crumbs leave spaces giving good drainage, aeration and easy root penetration; large flat platy peds pack together reducing all three (1)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	
06.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Fossil fuel technologies: secondary / tertiary recovery raising recovery from 20% to 60%; directional drilling and subsea wells; ROVs; fracking for tight oil and gas; tar sands / oil shales; coal gasification / liquefaction; enhanced gas recovery; methane hydrates; CCS (pre / post combustion, storage) • Limits: still finite (discoveries declining since the 1980s); rising energy and financial cost of unconventional sources; environmental impacts (aquifers, water use, tremors, land take, habitat damage); CO₂ and other pollution; CCS unproven at scale and impractical for vehicles • Nuclear: new uranium extraction (polymer adsorption from seawater, phosphate mining, coal ash); molten salt reactors; plutonium breeder and thorium reactors increasing fuel use many-fold; fusion as a long-term prospect • Limits: cost overruns, decommissioning, waste, public opposition, proliferation, embodied energy of low-grade uranium; fusion not commercial • Sustainability criteria: maintaining supply without unacceptable environmental, economic or social impact; comparison with renewables and energy conservation; energy gap argument • Judgement: technologies can extend supplies and reduce some impacts but cannot make non-renewables sustainable indefinitely; role as transition / base-load alongside renewables	9
06.2	<i>1 mark.</i> • 80-90% (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-196; AQA spec 3.3.3-3.3.4.1.	
07.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • No reservoir range = $3.9 - 0.7 = 3.2$ m; with reservoir = $2.7 - 1.7 = 1.0$ m (1) • Reduction = 2.2 m (1)	2

Question	Marking guidance	Marks
07.2	1 mark for each point. • After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream (1) • During dry periods the dam is opened to release water and maintain river levels (1)	2
07.3	1 mark for each point. • Deforestation / urbanisation reduce infiltration and interception (1) • So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods (1)	2
07.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Site factors: topography - narrow exit from a large deep basin so a small dam holds a large volume; impermeable, strong bedrock without faults or seismic activity; large catchment with regular high rainfall and low evaporation; catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation; forest / vegetation in the flooded area decaying anaerobically releasing methane; access for workers and materials and distance to demand; value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • Alternatives for supply: rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • Conservation strategies: low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • Evaluation: reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination has high energy / carbon cost; combination needed as demand grows • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	

Question	Marking guidance	Marks
08.1	<i>Correct answer only.</i> • C (1)	1
08.2	<i>1 mark for each point.</i> • Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms (1) • Tropospheric ozone is a secondary pollutant (from NO _x photochemistry) that is toxic to plants and humans and is a greenhouse gas (1)	2
08.3	<i>1 mark for each point.</i> • HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere (1) • So most do not persist long enough to reach the stratosphere where UV would release their chlorine (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
09.1	<i>1 mark for each point.</i> • Screens (metal grills) trap floating / suspended non-biodegradable items (paper, plastic) which are landfilled or incinerated (1) • Grit traps: the channel widens so flow slows and stones / road grit sink out because there is not enough kinetic energy to carry them (1)	2
09.2	<i>1 mark for each point.</i> • The effluent stands in large tanks so the faecal / organic solids sink and are removed as sludge (1) • This removes over 95% of the organic matter from the fluid so the risk of deoxygenation of the river is greatly reduced (1)	2
09.3	<i>1 mark for each point.</i> • Aerobic bacteria digest / break down the remaining dissolved and suspended organic matter (1) • Air (or pure oxygen) is mixed in by paddle wheels or air stones so oxygen shortage does not limit the aerobic process (1) • The suspended bacteria collected in the secondary sedimentation tank are the 'activated sludge'; returning them maintains a large bacterial population so digestion is rapid (1)	3
09.4	<i>1 mark for each point.</i> • Effluent from secondary treatment may still contain dissolved inorganic nutrients (phosphates, nitrates) that cause eutrophication, and some bacteria (1) • Needed where the discharge site is ecologically sensitive or used by many people (bathing water, drinking water abstraction) (1) • Phosphate stripping with iron(III) sulfate; micro-strainers (very fine sieves) then UV light or chlorine to kill remaining bacteria (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 269-272; AQA spec 3.4.3.2.9.	
10.1	<i>1 mark for the correct answer.</i> • $16.0 - 5.0 = £11$ million (1)	1

Question	Marking guidance	Marks
10.2	1 mark for each point. - It is rarely practical to reduce emissions to zero and the impact of low-level emissions may be acceptable, so an acceptable level must be decided (1) - ALARA: emissions should be As Low As Reasonably Achievable (1) - BATNEEC: the Best Available Technology Not Entailing Excessive Cost is chosen to achieve this (1)	3
10.3	1 mark for each point. - As industries expand or cities grow the number of pollutant sources increases (1) - So the total emissions / combined concentration rises even if each source is unchanged, and higher standards are needed (1)	2
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Critical Pathway Analysis: predicting movement of mobile persistent pollutants (state of effluent, wind, currents, geology, pH / oxygen and solubility, bioaccumulation, food sources, half-lives) so monitoring focuses on sites at risk; mainly radioactive discharges; checked by sampling - Critical Group Monitoring: exposure of the most-exposed group monitored before health effects; if their risk is acceptable everyone else is safer; emissions reduced if not - Emission control strategies: location (marine discharges, downwind of cities, not over aquifers) and timing (avoiding temperature inversions, low river flow) - Principles: polluter pays (incentive depends on penalties and enforcement), precautionary principle (assume harm until shown otherwise; ignorance no excuse), ALARA / BATNEEC and the non-linear cost-efficiency relationship - Selection of methods: production prevention (desulfurisation), prevention of release (precipitators, catalytic converters), post-release remediation (oil spill clean-up, phytoremediation), alternative processes (electric vehicles, renewables, pyrethroids) - Evaluation: prevention usually cheaper and more effective than remediation; costs rise steeply near 100%; CPA relies on understanding of pathways; enforcement and international agreement needed for transboundary pollutants; judgement about acceptable risk is political as well as scientific - Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	

Question	Marking guidance	Marks
11.1	Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Negative feedback: increased low-level cloud (albedo), increased photosynthesis and sequestration; hydrological cycle evaporation-precipitation • Positive feedback: melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • Tipping points: self-sustaining processes; why control must precede them; examples • Implications: urgency and precautionary principle; need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation; geoengineering risks • Difficulty of prediction because feedbacks interact at different scales; models; international agreements and per capita footprints • Judgement on whether current strategies are adequate Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Scale of the problem: how energy is used (transport, buildings, industry, water); links between affluence and per capita use • Transport: bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • Buildings: orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • Industry: heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • Limits: rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples; comparison with the cost of new supply; conclusion that conservation is the cheapest first step but cannot alone meet demand growth Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	25